MIL-P-48092B (AR)
5 April 1985
SUPERSEDING
MIL-P-48092A (AR)
2 June 1980

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE M509A1;
HARDWARE METALLIC FOR

This specification is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, provides quality assurance provisions, and packaging for the hardware, metallic for one type of projectile designated as Projectile, 8 Inch, HE, M509A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9241399 - Hardware, Metallic

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18-74 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (Dwg.) 9241399, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.4 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Cup, Expulsion Charge	9331648	25
Plate, Forward	9210124	25
Key	9210133	25
Key	9210134-1	25
Key	9210134-2	25
Key	9210135-1	25
Key	9210135-2	25
Key	9210137	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Metallic Hardware				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, Expulsion Charge (9331648)				
	Examination for Defects	25	-	3.2	4.4.2.1
	Hardness Test (See note a.)	15	-	3.2	4.5.2
	Plate Forward (9210124)				
	Examination for Defects	25	-	3.2	4.4.2.2
	Key (9210133)				
	Examination for Defects	25	-	3.2	4.4.2.3
	Key (9210134-1)				
	Examination for Defects	25	-	3.2	4.4.2.4
	Key (9210134-2)				
	Examination for Defects	25	-	3.2	4.4.2.4
	Key (9210135-1)				
	Examination for Defects	25	-	3.2	4.4.2.5
	Key (9210135-2)				
	Examination for Defects	25	-	3.2	4.4.2.5
	Key (9210137)				
	Examination for Defects	25	-	3.2	4.4.2.6

NOTES:

a. Applicable only when alternate material is used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331648
4.4.2.1	Cup, Expulsion Charge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread		.40%	3.2	Gage
102	Major diameter of thread		.40%	3.2	Gage
103	True position of largest inside diameter with pitch diameter of threads		.40%	3.2	Gage
104	True position of smallest inside diameter with pitch diameter of thread		.40%	3.2	Gage
105	Minimum inside diameter		.40%	3.2	Gage
106	True position of largest outside diameter with pitch diameter of thread		.40%	3.2	Gage
107	Length of threaded area		.40%	3.2	Gage
108	Hardness test (only applies for alternate material)	10	Note a	3.2	4.5.2
109	Thickness of bottom		.40%	3.2	Gage
110	Thickness of sidewall		.40%	3.2	Gage
111	Overall length		.40%	3.2	Gage
112	Depth of counterbore		.40%	3.2	Gage
113	Presence of two slots		.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:
a. Each heat treat batch (see 6.5). If any cup fails to comply with the applicable requirements, the heat treat batch represented by the sample shall be rejected.

DRSMC (D) Form 160, 1 Aug 83 replaces edition of 1 Ju[illegible] which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9210124
4.4.2.2	Plate, Forward				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of packing groove	See 4.5.1	.40%	3.2	Gage
102	Width of packing groove		.40%	3.2	Gage
103	Outside diameter		.40%	3.2	Gage
104	Chamfer missing		.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092B (AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.3	Key		1 OF 1		9210133
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of dovetail at basic dimension from shoulder		.40%	3.2	Gage
102	Length of dovetail portion		.40%	3.2	Gage
103	Height of dovetail max.		.40%	3.2	Gage
104	Distance from shoulder to concave section at basic width		.40%	3.2	Gage
105	Total width, maximum		.40%	3.2	Gage
106	Overall length		.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Key				9210134-1 and -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of dovetail to basic distance from shoulder		.40%	3.2	Gage
102	Height of dovetail		.40%	3.2	Gage
103	Distance from shoulder to concave section at basic width		.40%	3.2	Gage
104	Total width, maximum		.40%	3.2	Gage
105	Dimension "X" improper		.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48092B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9210135-1 and -2
4.4.2.5	Key				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of dovetail at basic distance from shoulder		.40%	3.2	Gage
102	Height of dovetail		.40%	3.2	Gage
103	Distance from shoulder to concave section at smaller basic width		.40%	3.2	Gage
104	Distance from shoulder to concave section at larger basic width		.40%	3.2	Gage
105	Total width		.40%	3.2	Gage
106	Dimension "X" improper		.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9210137
4.4.2.6	Key				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of dovetail at basic distance from shoulder		.40%	3.2	Gage
102	Height of dovetail		.40%	3.2	Gage
103	Distance from shoulder to concave section at smaller basic width		.40%	3.2	Gage
104	Distance from shoulder to concave section at larger basic width		.40%	3.2	Gage
105	Total Width		.40%	3.2	Gage
106	Length of dovetail portion		.40%	3.2	Gage
107	Overall length		.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.4	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Hardness test of cup. The cups selected shall be subjected to the hardness test in accordance with the method and procedures of ASTM E 18-67. The load and penetrator shall be in accordance with the hardness scale specified on the applicable drawing. Three readings shall be taken on each cup on the outside surfaces of the closed end approximately 1/4" from the edge and approximately 120° apart. A suitable fixture shall be utilized to provide support under the surface being tested. Such fixtures shall be approved in accordance with 4.4.4. If the average value of the three readings for any cup is less than the value specified in the applicable drawing, that cup shall have failed the requirement. Equipment used for performing this test shall be approved in accordance with paragraph 4.4.4.

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5. PACKAGING

5.1 Preservation.

5.1.1 Level B. The metallic hardware shall be preserved and packaged in accordance with MIL-STD-1169. The bulk packaging arrangement shall not be utilized. The expulsion charge cup (Dwg. 9331648) shall be packaged utilizing the single item packaging arrangement.

5.2 Packing.

5.2.1 Level B. The metallic hardware preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169. Each pack shall contain only one part number.

5.3 Marking. Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the Projectile, 8 Inch, HE, M509A1.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Center (ARDC), ATTN: AMSMC-QAR-I (D), Dover, NJ 07801-5001.

6.4 Submission of results of contractor-conducted examinations and tests. Data shall be submitted in accordance with data item DI-R-1721 which is listed on the DD Form 1423 for the contract.

6.5 Heat-treatment batch. A heat-treatment batch is herein defined as cups, of the same type, that have been produced from one lot of material, and heat treated at the same time in the same furnace, quench medium and aging process for each of all phases of the entire heat treatment process. When a continuous-type heat treatment process is used, the quantity of cups processed through the furnace for each four (4) hours production, or portion thereof, shall be considered as a heat treatment batch.

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6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additons, modifictions, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on he entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48092B	PROJECTILE, 8 INCH, HE M509A1; HARDWARE METALLIC FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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